

RESEARCH ARTICLE

Optimization of an allelic discrimination real-time RT-PCR assay for detection of H275Y oseltamivir resistance gene mutation among influenza A(H1N1)pdm09 patients from 2020 to 2022

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Abstract

Influenza virus is known to cause mild to severe respiratory infections and is also prone to genetic mutations. Of all the mutations, neuraminidase (NA) gene mutations are a matter of concern, as most approved antivirals target this protein. During the 2020 influenza season, an emergence of mutation in the NA gene, affecting the binding of the World Health Organization (WHO)-recommended probes to the specific site of the NA gene, was reported by our group. As a result of this mutation, the WHO-recommended allelic discrimination real-time reverse transcriptase polymerase chain reaction (RT-PCR) assay was unable to detect wild-type (H275) or mutant oseltamivir-resistant (Y275) strains of influenza A(H1N1)pdm09 viruses. In the current study, the WHO-recommended probes were redesigned according to the mutation in the probe binding site. Fifty undetermined samples (2020–2021) from the previous study were retested with the newly designed probes and found to be positive for H275 and/or Y275. The results obtained were similar to the Sanger sequencing results from the previous study, suggesting that the redesigned probes were efficient in discriminating between wild-type and mutant-type viruses. Furthermore, 133 samples from 2022, making a total of 183 samples (2020–2022), were tested using improved allelic discrimination real-time RT-PCR, and the overall prevalence rate of oseltamivir resistance in 2020–2022 was found to be 0.54%.

KEYWORDS

H275Y mutation, improved real-time allelic discrimination RT-PCR, influenza A(H1N1)pdm09 virus, oseltamivir resistance, redesigned probes, single-nucleotide polymorphism (SNP)

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1 | INTRODUCTION

Influenza virus is a respiratory pathogen transmitted through droplets or by direct contact.¹ Seasonal flu epidemics occur worldwide almost every year, most commonly during the months of winter. However, different types of influenza viruses vary widely in severity and extent of spread.² Influenza A viruses are responsible for global pandemics that, although unpredictable, may occur every 10–40 years based on the four pandemics that emerged over the past 100 years.³ Currently circulating influenza A(H1N1)pmd09 virus is a subtype of influenza A that caused the pandemic in 2009.⁴

Neuraminidase inhibitors (NAIs), such as zanamivir, oseltamivir, peramivir, and baloxavir marboxil, are the antivirals used to treat influenza infection.⁵ The NAIs, namely oseltamivir, continue to represent the most widely used antivirals for the management of influenza virus infection.⁶ Oseltamivir is a prodrug of oseltamivir carboxylate, which was approved as an antiviral drug in 1999. Any mutation in the NA gene of influenza viruses, particularly those located at the drug binding site, can lower the ability of oseltamivir to inhibit the enzymatic activity of NA, causing oseltamivir resistance or reduced susceptibility.⁷ A mutation known as H275Y in the NA gene confers oseltamivir resistance in the influenza A (H1N1)pmd09 virus.⁶ The H275Y mutation involves a nucleotide change from cytosine to thymine (C to T) at position 823, resulting in the amino acid substitution of histidine to tyrosine (H to Y) at position 275, leading to a conformational change in amino acids Glu276 and Arg224 that is required for oseltamivir to bind and function.^{7,8} Viruses detected with the H275Y amino acid substitution are clinically relevant due to the frequency of their prevalence and availability of supporting clinical data demonstrating reduced treatment efficacy with oseltamivir.⁹

The World Health Organization (WHO) recommends allelic discrimination real-time reverse transcriptase polymerase chain reaction (RT-PCR) assay as a rapid method for detection of the H275Y mutation. In 2022, our study on the detection of oseltamivir resistance using the WHO-recommended single-nucleotide polymorphism (SNP) assay reported the presence of a mutation of guanine to adenine (G to A) at the probe binding site, position 831, of the NA gene sequence of influenza A(H1N1)pmd09 virus samples from 2020 onwards. This affected the binding of the WHO-recommended probes to a specific site of the NA gene, resulting in the inability to detect the H275Y mutation using the WHO-recommended allelic discrimination real-time RT-PCR assay.¹⁰

Thus, the aim of the current study was to redesign the probes according to the mutation at the probe binding site to detect the H275Y mutation and estimate the prevalence of oseltamivir-resistant influenza A(H1N1)pmd09 virus infection among patients from 2020 to 2022.

2 | MATERIALS AND METHODS

2.1 | Study population

A laboratory-based retrospective study was carried out at the Manipal Institute of Virology (MIV) using throat and/or nasopharyngeal swab samples archived from January 2020 to December

2022. A total of 183 influenza A(H1N1)pmd09 virus-positive samples—50 from the previous study (2020–2021)¹⁰ and 133 from 2022—with a C_t value < 30 were selected by purposive sampling and included in the study.

2.2 | Probe redesigning

To redesign the probes, the NA gene sequences of influenza A(H1N1)pmd09 virus from 2020 to 2021, which were generated as part of a previous study, were obtained from the GenBank database (OP737518 to OP737538, OQ363464 to OQ363493). Also, genome sequences of influenza A(H1N1)pmd09 vaccine strains such as A/Victoria/2570/2019 (EPI_ISL_417210) and A/Wisconsin/588/2019 (EPI_ISL_404460) obtained from Global Initiative on Sharing All Influenza Data (GISAID) were downloaded and used as references. Sequences were aligned, and mutations in the probe binding site at position 831 were identified. Probes were redesigned according to the mutation in the probe binding site by replacing nucleotide guanine with adenine (G with A) at position 831. The binding of the redesigned probes to the NA gene was verified using MEGA X version 11 software (Figure 1A).

2.3 | Optimization of allelic discrimination real-time RT-PCR

The FavorPrep™ Viral DNA/RNA kit (FAVORGEN; Biotech Corp.) was used to extract nucleic acids from the samples selected for the study. An improved allelic discrimination real-time RT-PCR assay was performed using NA gene-specific WHO-recommended primers from Applied Biosystems and redesigned probes were obtained from Eurofins Genomics. A QuantiTect virus kit (Qiagen) was used to perform improved allelic discrimination real-time RT-PCR. Nucleic acid extracted from a standard strain of influenza A/Victoria/2570/2019(H1N1)pdm09-like virus received from the National Institute for Biological Standards and Control (NIBSC; No. 21/346) was used as a wild-type virus (H275) positive control. For the mutant-type virus (Y275), a synthetic positive control was designed using the NA sequence of the 2021–2022 vaccine reference strain by modifying the probe binding site at position 823.

The assay components used were 25 µL of reaction mix containing 5 µL of 5X Master mix, 0.2 µL of RT Mix, 1 µL of 10 µM concentration of each probe (H275 and Y275), 1 µL of 2 µM concentration of both primers, 10.8 µL of nuclease-free water and 5 µL of template RNA. Cycling conditions were pre-read for 1 min at 60°C, reverse transcription for 20 min at 50°C, initial denaturation for 10 min at 95°C, 45 cycles of amplification (denaturation for 15 s at 95°C, annealing and extension for 45 s at 56°C), and finally post-read for 1 min at 60°C. The PCR run was performed using Applied Biosystems QuantStudio™ 5 and 6 Flex Real-Time PCR Systems. Endpoint fluorescent results of samples were analyzed, and an allelic discrimination plot was generated.

(A)

1. WHO Recommended H275 Probe	-----C A C T A T G A G G A A T-----
2. WHO Recommended Y275 Probe	-----T A C T A T G A G G A A T-----
3. A/India-Karnataka/MIVAN3046/2020(H1N1)	A T G A A A G C C C C T A A T T A T C A C T A T G A A G A A T G C T C C T G T T A C C C T G A T T C T
4. A/India-Karnataka/MIVAN3536/2020(H1N1)	A T G A A A G C C C C T A A T T A T C A C T A T G A A G A A T G C T C C T G T T A C C C T G A T T C T
5. A/India-Karnataka/MIVAB8366/2021(H1N1)	A T G A A A G C C C C T A A T T A T C A C T A T G A A G A A T G C T C C T G T T A C C C T G A T T C T
6. A/India-Karnataka/MIVAB9332/2021(H1N1)	A T G A A A G C C C C T A A T T A T C A C T A T G A A G A A T G C T C C T G T T A C C C T G A T T C T
7. Influenza A virus (A/Victoria/2570/2019(H1N1))	A T G A A A G C C C C T A A T T A T C A C T A T G A A G A A T G C T C C T G T T A C C C T G A T T C T
8. Influenza A virus (A/Wisconsin/588/2019(H1N1))	A T G A A A G C C C C T A A T T A T C A C T A T G A A G A A T G C T C C T G T T A C C C T G A T T C T
9. Redesigned H275 Probe	-----C A C T A T G A A G A A T-----
10. Redesigned Y275 Probe	-----T A C T A T G A A G A A T-----

(B)

Improved allelic discrimination real time RT-PCR results

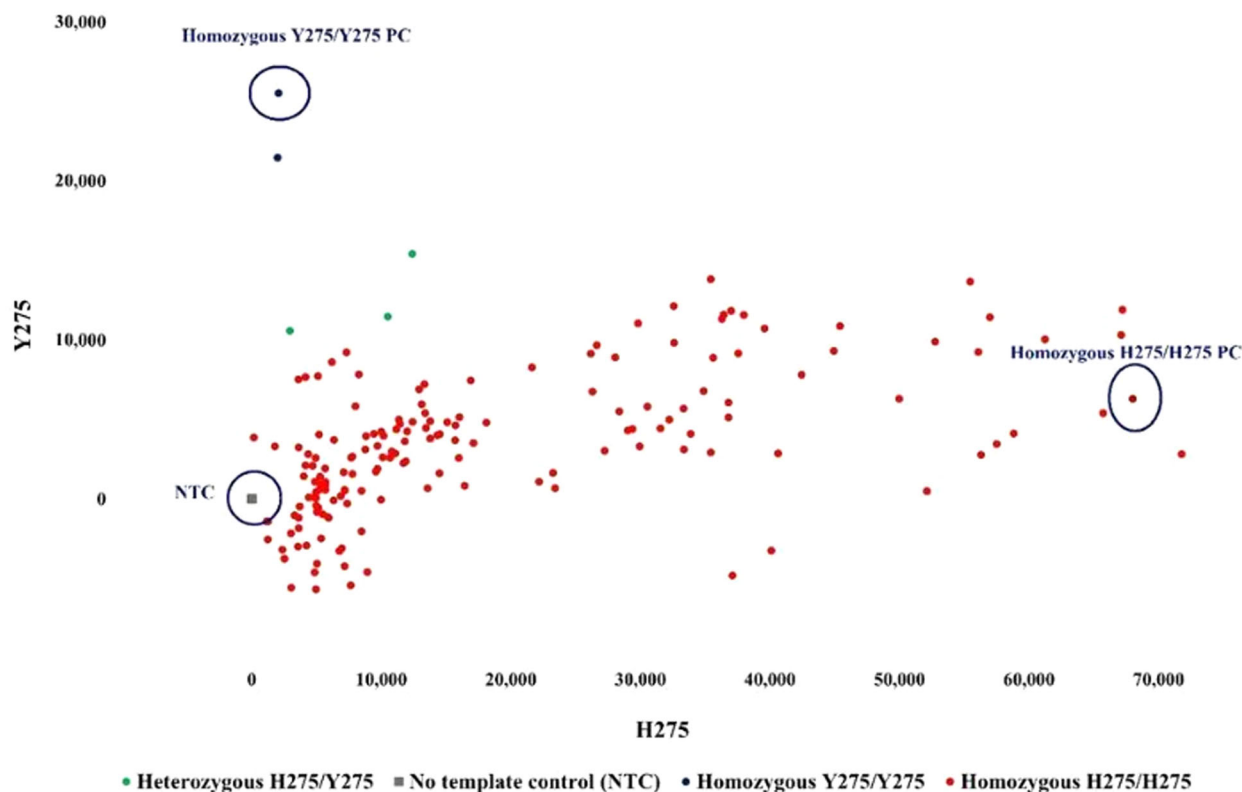


FIGURE 1 Allelic discrimination assay for detection of H275Y oseltamivir resistance gene mutation. (A) Multiple sequence alignment of World Health Organization (WHO)-recommended and redesigned probes along with genome sequences of the samples and WHO-recommended vaccine strains. Highlighted region shows position 831. (B) Results of improved allelic discrimination real-time reverse transcriptase polymerase chain reaction (RT-PCR). The red dots on the X-axis indicate the samples that are homozygous wild-type H275/H275; blue dots on the Y-axis indicate the sample that is homozygous mutant-type Y275/Y275; green dots indicate the samples that were heterozygous type H275/Y275; points encircled indicate the homozygous H275 and Y275 positive controls. Grey square indicate no template control (NTC).

2.4 | Sanger sequencing

The samples that were detected as heterozygous (H275 and Y275) by improved allelic discrimination real-time RT-PCR were confirmed with Sanger sequencing by amplifying the NA segment using NA gene-specific WHO-recommended primers and the SuperScript™ III One-Step RT-PCR System (Invitrogen). The desired product size of 620 bp was separated using a 0.8% agarose gel, excised by a gel extractor, and purified using a GenElute™ Gel Extraction Kit (Sigma-Aldrich). The nucleic acid concentration of the purified gel product

was quantified by a Nanodrop™ 2000, sequenced using Applied Biosystems Big Dye Terminator Cycle sequencing kit (version 3.1), and analyzed using a 3500XL Genetic analyzer (Applied Biosystems).

2.5 | Sequence analysis

The sample sequences and the reference sequences of influenza A(H1N1)pdm09 vaccine strains A/Victoria/2570/2019(H1N1)pdm09-like virus (EPI_ISL_417210; egg-based vaccine) and of influenza

A/Wisconsin/588/2019(H1N1)pdm09-like virus (EPI_ISL_404460; cell-culture-based vaccine) from GISAID were aligned using Mega X version 11 software.

To detect the global prevalence of guanine to adenine (G to A) mutation at position 831 of the NA gene sequence from 2020 onwards, NA gene sequences from different continents, namely Africa, Asia, Europe, North America, Oceania, South America, and Antarctica, were downloaded from GISAID and analyzed. A total of 16095 NA sequences from 2020 to 2022 were analyzed for mutation G831A.

3 | RESULTS

3.1 | Improved allelic discrimination real-time RT-PCR

Improved allelic discrimination real-time RT-PCR was performed with redesigned probes for 183 samples that included 50 samples from our previous study (2020–2021).¹⁰ Of the 50 samples, 48 were homozygous wild-type (H275/H275), one was homozygous mutant-type (Y275/Y275), and one was heterozygous type (H275 and Y275; Table 1). The sample that was detected to be heterozygous type was confirmed to be homozygous wild-type by Sanger sequencing in the previous study.¹⁰ However, the sample was resequenced to further confirm the results of allelic discrimination PCR.

Furthermore, 133 samples from 2022 were tested, and the results were analyzed and classified as homozygous wild-type (H275/H275), mutant-type (Y275/Y275), and heterozygous type (H275 and Y275). Of 183 samples in total, 179 samples were positive for homozygous wild type (H275/H275), three samples were positive for heterozygous type (H275 and Y275), and one sample was positive for homozygous mutant-type (Y275/Y275) (Figure 1B, Table 1).

3.2 | Sequence analysis

The sample from 2020 that was detected to be heterozygous by improved allelic discrimination real-time RT-PCR was confirmed to be

negative for the mutation by Sanger sequencing in both previous (OQ363479) and current study.

Sequence analysis of two heterozygous-type (OR405270, OR405271) samples from 2022 in comparison with the reference sequence of influenza A/Wisconsin/588/2019(H1N1)pdm09-like virus confirmed the absence of nucleotide substitution of cytosine to thymine (C to T) at position 823 that resulted in the H275Y mutation and was confirmed as wild-type virus.

Analysis of NA sequences of influenza A(H1N1)pmd09 viruses from the different continents revealed that percentage prevalence of G831A showed an upward trend from 2020 onwards with the percentage of sequences with mutation (A831) increasing from 37.16% to 98.27% during 2020–2022 (Figure 2, Table 2).

4 | DISCUSSION

The H275Y amino acid mutation among influenza A(H1N1)pmd09 viruses has a worldwide prevalence and is clinically important because it can compromise the effectiveness of oseltamivir and impact the ability to manage and control influenza outbreaks. Continuous monitoring of H275Y mutation in circulating influenza A(H1N1)pmd09 virus is essential since these substitutions are the only ones consistently associated with the clinical resistance that occurs with sufficient frequency globally. Currently, NA sequencing and allelic discrimination real-time RT-PCR assay are helpful genotypic methods for detecting H275Y associated with oseltamivir resistance.⁹ allelic discrimination Real-time RT-PCR is one of the methods recommended by WHO for detecting SNPs (H275Y) and is found to be an ideal method for detecting SNPs while handling a large number of clinical isolates since it is a simple and rapid genotypic method with a lower risk of cross-contamination.^{11,12} However, mutations that occur commonly in influenza viruses reduce the effectiveness of allelic discrimination real-time RT-PCR in detecting the H275Y mutation. Our study on oseltamivir resistance published in 2022 reported the presence of a mutation in the probe binding site at position 831 of the NA gene sequence of the influenza A(H1N1)pmd09 virus from 2020 onwards. This mutation affects the

TABLE 1 Comparison of improved allelic discrimination real-time RT-PCR results with Sanger sequencing results from the previous and current study.

Previous study ¹⁰					Current study		
Year	No. of samples	WHO allelic discrimination real-time RT-PCR	Sanger sequencing		Improved allelic discrimination real-time RT-PCR		
			Homozygous wild type (H275)	Homozygous mutant type (Y275)	Homozygous wild type (H275)	Homozygous mutant type (Y275)	Heterozygous type (H275/Y275)
2020	38	Undetermined	37 (97.4%)	1 (2.6%)	36 (94.7%)	1 (2.6%)	1 (2.6%)
2021	12	Undetermined	12 (100%)	0	12 (100%)	0	0
2022	133	-	-	-	131 (98.5%)	0	2 (1.5%)
Total	183		49	1	179	1	3

FIGURE 2 Sequence analysis of neuraminidase (NA) gene of influenza A(H1N1)pmd09 viruses from different continents (2020–2022). X-axis represents year and continents; Y-axis represents the number of NA sequences shown as bars with mutation (blue) and without mutation (orange).

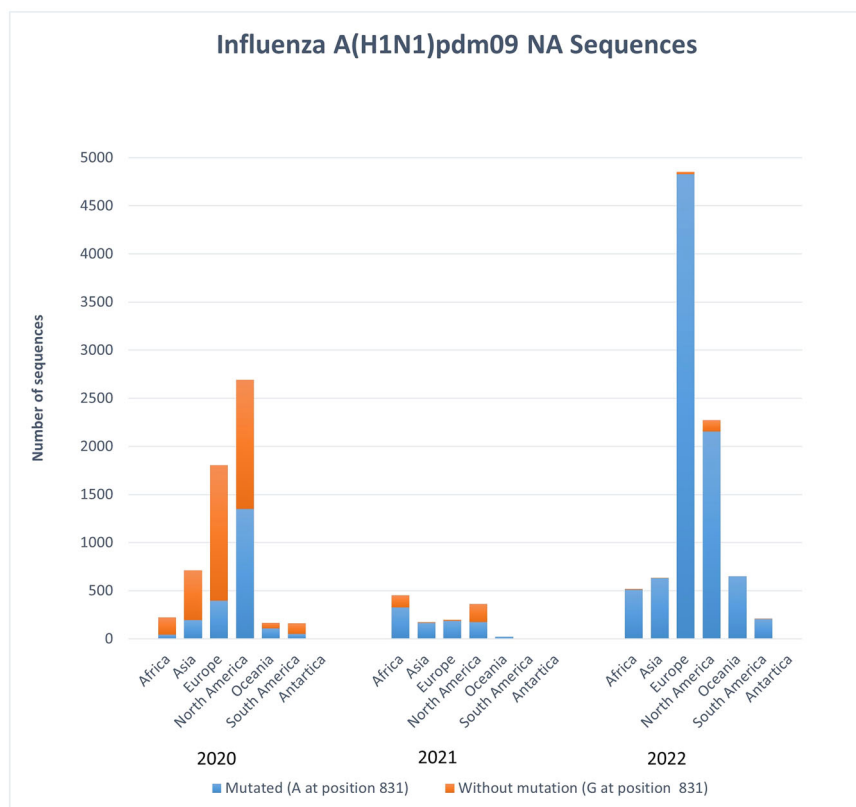


TABLE 2 Number and percentage of influenza A(H1N1)pmd09 gene sequences with mutation (G to A) at position 831 from 2020 to 2022.

Number of influenza A(H1N1)pmd09 NA gene sequences									
Continents	2020			2021			2022		
	A831	G831	Mutation (%)	A831	G831	Mutation (%)	A831	G831	Mutation (%)
Africa	42	180	18.91	327	124	72.50	510	9	98.26
Asia	196	515	27.56	167	8	95.42	631	3	99.52
Europe	394	1409	21.85	185	14	92.96	4826	26	99.46
North America	1347	1343	50.07	172	189	47.64	2154	118	94.80
Oceania	108	57	65.45	22	0	100	648	0	100
South America	51	110	31.67	0	1	0	207	2	99.04
Antarctica	0	0	0	0	0	0	0	0	0
Total	2138	3614	37.16	873	336	72.2	8976	158	98.27

binding of WHO-recommended probes to the specific site on the NA gene to detect the H275Y mutation, resulting in undetermined assay results.¹⁰

In the present study, the WHO-recommended probes were redesigned according to the mutation in the probe binding site at position 831. The binding of the redesigned probes to the NA gene was confirmed. A total of 50 samples from 2020 to 2021 that were undetermined using the WHO-recommended allelic discrimination real-time RT-PCR assay in a previous study¹⁰ were retested by improved allelic discrimination real-time RT-PCR using redesigned probes, and all the samples tested positive for H275 and/or Y275.

Out of 50 samples tested, the sample that was confirmed to have a mutation by Sanger sequencing (OQ363483) in our previous study¹⁰ was detected to be homozygous mutant type by improved allelic discrimination real-time RT-PCR. One sample that was confirmed as wild-type by Sanger sequencing in our previous study was found to be heterozygous with improved allelic discrimination real-time RT-PCR. The sample was resequenced by Sanger sequencing to confirm that it does not have a “mixed” population of both H275 and Y275 viruses. However, the results of Sanger sequencing were similar to the previous results. Accurate detection of heterozygous or mixed populations of viruses is difficult with Sanger sequencing since

Sanger sequencing identifies the predominant gene sequence among the mixed population of viruses. A previous study reported that the limit of detection of Sanger sequencing for “mixed population” of wild-type (H275) and mutant-type (Y275) ranges from 10% to 25%. Further, the study also reported pyrosequencing method for detection-resistant viruses in “mixed population” at levels as low as 10%.¹³ However, pyrosequencing is not a widely used method and needs controls for validation.⁹ Thus, samples determined to be heterozygous in allelic discrimination PCR may need to be analyzed using newer technologies such as next-generation sequencing (NGS) or digital PCR to detect mutations in the mixed viral population. However, these methods are expensive and not within the scope of this study.¹³ The overall results of improved allelic discrimination real-time RT-PCR were 98% similar to the results of Sanger sequencing of the previous study. Thus, the redesigned probes were found to be efficient in discriminating between wild-type viruses and mutant viruses.

Global prevalence of G831A mutation in NA sequences of influenza A(H1N1)pmd09 viruses was also analyzed as part of the study. Sequence analysis showed that global prevalence of G831A mutation has been increasing since the year 2020. In 2022, 98.27% of NA sequences were detected with the mutation. These observations further strengthen our findings and recommend revision of WHO-recommended SNP assay probes to detect the H275Y mutation.

The current study also estimated the prevalence of oseltamivir resistance using newly designed probes, followed by confirmation of gene mutations using Sanger sequencing. Among a total of 183 samples, 182 were homozygous wild type (H275/H275), and one sample was homozygous mutant type (Y275/Y275). The overall prevalence rate of oseltamivir resistance from 2020 to 2022 was calculated to be 0.54%. The prevalence rate of oseltamivir resistance was similar to that in the previous studies from India that reported resistance rates of 0.15% (2015–2016)¹⁴ and 0.27% (2014–2021).¹⁰ Further, global studies reported similar resistance rates of 0.5% during 2014–2015,¹⁵ 0.8% during 2015–2016,¹⁶ 0.2% during 2016–2017,¹⁷ 0.8% during 2017–2018,¹⁸ and 0.6% during 2018–2020.¹⁹ The lower prevalence rate of oseltamivir resistance suggests that oseltamivir can be continued as an antiviral drug to treat influenza.

5 | CONCLUSION

This study presents the importance of probe redesigning and using redesigned probes in optimi allelic discrimination real-time RT-PCR to detect the oseltamivir-resistant influenza A(H1N1)pmd09 virus and suggests considering the revision of the WHO-recommended SNP assay to detect the H275Y mutation.

This study further reiterates our previous observation that the prevalence rate of oseltamivir resistance is low in the Indian population and suggests the continuation of oseltamivir carboxylate as an antiviral drug against influenza.

AUTHOR CONTRIBUTIONS

Anitha Jagadesh conceived and guided the research work. Gandhapu Harish performed all the experiments and data analysis. Ujwal Shetty contributed to performing Sanger sequencing. Prasad Varamballi contributed to the data analysis. Chiranjay Mukhopadhyay provided insightful discussion.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data related to the current study are available from the corresponding author upon request.

ETHICS STATEMENT

The study was approved by the Institutional Ethics Committee (IEC), Kasturba Medical College, MAHE (DHE Registration no. EC/NEW/INST/2021/1707) (Approval IEC: 623/2022).

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